Lab Manual: Filesystem Surgery (MBR & Partition Recovery)

1. Aim

To simulate MBR corruption and recover the partition table using TestDisk while preserving the existing filesystem and data.

2. Tools Required

Linux system (Kali / Ubuntu)

dd

losetup

fdisk

testdisk

mkfs.ext4

Install TestDisk if not installed:

sudo apt update

sudo apt install testdisk

3. Lab Setup Architecture

Disk Image → Loop Device → Partition → EXT4 Filesystem → Secret File

4. Step 1: Clear Existing Loop Devices

sudo losetup -D

5. Step 2: Create a Virtual Disk Image

dd if=/dev/zero of=disklab.img bs=1M count=50

6. Step 3: Attach Disk Image as Loop Device

sudo losetup -Pf disklab.img

losetup -a

7. Step 4: Create a Partition

sudo fdisk /dev/loop0

Inside fdisk:

n

p

1

Enter

Enter

w

8. Step 5: Reload Partition Table

sudo partprobe /dev/loop0

ls /dev/loop0*

Expected:

/dev/loop0

/dev/loop0p1

9. Step 6: Format the Partition

sudo mkfs.ext4 /dev/loop0p1

10. Step 7: Mount the Partition

sudo mkdir -p /mnt/disklab

sudo mount /dev/loop0p1 /mnt/disklab

11. Step 8: Create Secret Data

sudo mkdir /mnt/disklab/SECRET_DATA

echo "IIITB_2026" | sudo tee /mnt/disklab/SECRET_DATA/key.txt

ls /mnt/disklab/SECRET_DATA

12. Step 9: Unmount and Detach

sudo umount /mnt/disklab

sudo losetup -D

13. Step 10: Sabotage the Disk (Delete MBR)

dd if=/dev/zero of=disklab.img bs=512 count=1 conv=notrunc

14. Expected Output

sudo mount disklab.img /mnt/disklab

Error:

mount: unknown filesystem type

Reason: Linux cannot locate the partition table.

15. Recovery Process Using TestDisk

sudo testdisk disklab.img

Inside TestDisk select:

Proceed

None

Analyse

Quick Search

Write

16. Reload the Recovered Disk

sudo losetup -Pf disklab.img

17. Mount the Recovered Partition

sudo mount /dev/loop0p1 /mnt/disklab

18. Verify Recovered Data

cat /mnt/disklab/SECRET_DATA/key.txt

Expected Output:

IIITB_2026